Elijah Andrae

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Curriculum Vitae | Semiconductor process, metrology, and statistical modeling

PROFILE

Data science graduate (B.S., University of Oregon, 2026; Mathematics and Business minors) who earned the degree doing semiconductor process work in a cleanroom. Wrote 26 cleanroom SOPs, each checked off by lab staff, ran a term-long designed-experiment program in JMP, and spent a year designing, fabricating, and characterizing microfluidic electrochemical devices. On the data side, builds simulations and machine learning models from large, imperfect records and tests them on held-out data. US citizen; ready to relocate.

EDUCATION

University of Oregon
2022 - 2026

B.S. Data Science (Marketing Analytics concentration); Minors: Mathematics, Business Administration

- Dean's List: Spring 2025, Fall 2025
- Data science: Machine Learning for Data Science; Probability and Statistics for Data Science; Principles and Techniques of Data Science; Data Structures in Python; Data Science for Social Justice; Foundations of Data Science I and II
- Mathematics and statistics: Stochastic Processes; Introduction to Mathematical Methods of Statistics I; Elementary Linear Algebra I and II; Several-Variable Calculus I and II; Introduction to Differential Equations; Introduction to Mathematical Cryptography; Introduction to Proof
- Physics and nanofabrication: Nanofabrication (A+); Research: Quantum and Nanotech (microfluidic device research); Research (independent electronics and optics laboratory sequences)
- Business and analytics: Marketing Analytics; Marketing Research; Finance; Accounting; Excel data analysis

Cathedral Preparatory School
2018 - 2022

High school; Erie, PA

RESEARCH EXPERIENCE

Electrochemical Microbubble Devices: Fabrication, Characterization, and Stochastic Signal Applications
Mar 2026

Independent Research, Alemán Lab, University of Oregon

- Designed and fabricated microfluidic electrochemical devices across two prototype generations in a year-long independent research project.
- Drew the photomask sets for both generations in CleWin, defining electrode geometry, gap spacing, interlayer alignment marks, and bond-pad routing on a 25 mm mask frame; the second generation addresses seven measurement channels through a shared-electrode matrix from a reduced pad count.
- Fabricated six working prototypes end to end: photolithography, thin-film metal deposition and lift-off, dielectric insulation, SU-8 molds, and PDMS channel integration on glass.
- Characterized activation thresholds, current response, field and gap dependence, and trial-to-trial variability on a Keithley 2450 and semiconductor probe station.
- Automated synchronized acquisition in Python (SCPI instrument control, logging, structured datasets): 80+ test runs across about 20 devices on four chips.
- Derived an activation model from reaction kinetics and modeled the first-event time as a stochastic process, using censored maximum likelihood so trials with no event stayed in the data.

Tools: CleWin, KLayout, Photomask design, Photolithography, LaserWriter, Thin-film deposition, E-beam deposition, PDMS soft lithography, Keithley 2450, Semiconductor probe station, Optical microscopy, Python

Emergency Dispatch Capacity Simulation
Sep 2026

Independent research extending DSCI 410L coursework (in progress)

- Built a gray-box discrete-event simulation from about 1.2 million public dispatch records (Eugene, 2017 to 2025) to estimate what added crisis-van capacity would have bought: one more van from 11 a.m. to 10 p.m. cuts the routine median wait by about half and closings without a van by about a third.
- Audited the record before modeling: a call's close time does not mark when the van came free, unsent requests are invisible before May 2022, and only a call's lead unit is named.
- Estimated quantities the record never logged, rebuilding van shifts and van-free times from the sequence of dispatches, and fitted six behavior settings jointly on May 2022 to 2023, testing on 2024 (typical miss 16 to 21% per measure).
- Ranked model mechanisms with a 16-run resolution IV fractional factorial design run on the simulator.
- Tested learned components against the hand-built dispatch rule: gradient-boosted trees and a PyTorch network trained on about 214,000 free-van minutes reached held-out AUC 0.880 and 0.877 versus 0.871; decoded, the network learned the same rule.
- Packaged the work as a Python codebase of about 19 modules with cached run scripts and a LaTeX paper build with automated number checks.

Tools: Python, discrete-event simulation, Markov chains, censored MLE, PyTorch, gradient-boosted trees, LaTeX

CLEANROOM AND LABORATORY EXPERIENCE

Nanofabrication Cleanroom Labs (PHYS 495) - DOE-Driven Process Development

Sep 2025 - Dec 2025

PHYS 495 Nanofabrication, University of Oregon

- Wrote 26 standard operating procedures, each checked off by lab staff, qualifying for independent use of furnace, deposition, lithography, etch, and metrology tools in an ISO 6 (Class 1000) cleanroom; also trained to enter an ISO 4 (Class 10) cleanroom.
- Operated a Lindberg tube furnace, Laurell spin coater, two Angstrom Covap thermal evaporators, Suss MJB4 mask aligner, Microtech LaserWriter (405 nm direct write), Novascan UV-ozone, March O2 plasma etcher, and wet etch under full gowning.
- Designed, ran, or analyzed four factorial experiments (two 2^4, two 2^3) plus a 12-level exposure dose test and a three-lens dose profile, about 60 designed runs modeled in JMP with reduced models, Box-Cox transforms, residual diagnostics, and desirability optimization.
- Identified the evaporator tool, not the recipe, as the largest source of thickness variation in a pooled class model; suspected crystal-monitor drift on one tool, reported it to lab staff, and recommended logging the in-situ monitor on every run.
- Cross-checked thickness metrology across two reflectometers, an ellipsometer, and a stylus profilometer: the two reflectometers showed no difference ($p = 0.999$) and the ellipsometer tracked the F40 at $r = 0.9999$.
- Switched dose-test measurement to the profilometer after a matched-pairs check showed reflectometer data too noisy, and recommended a 115 mJ/cm² exposure dose; found LaserWriter lens choice significant (ANOVA $p < 0.0001$).
- Showed in a 2^4 spin-coat screening design that final spin speed dominated resist thickness and acceleration drove coverage.
- Compared lift-off and wet-etch pattern transfer at about 97 nm metal thickness and aligned two- and three-layer patterns on mask-aligner fiducials and direct-write alignment marks.

Optical Systems & Interferometry Laboratory

Mar 2026 - Jun 2026

Independent Laboratory Sequence, 16 stations, University of Oregon

- Built and aligned free-space optical systems across a 16-station HeNe laboratory sequence, including a beam expander, optical isolators, a Michelson interferometer, and a scanning confocal Fabry-Perot cavity.
- Measured refractive index by Brewster's angle (about 1.54) and by Michelson fringe counting, and measured laser coherence length (about 15 cm).
- Built the Fabry-Perot cavity from 10 cm focal-length mirrors (free spectral range 375 MHz) and calculated a theoretical resolution of 12.6 MHz.
- Profiled the beam by pinhole scan, measured polarization states and photodiode rise times, and cross-checked optical power between a power meter and a photodiode method (within 13%).

Electronics Laboratory

Jan 2026 - Mar 2026

Independent Laboratory Sequence, University of Oregon

- Built and measured passive, diode, transistor, op-amp, and optoelectronic circuits across a 12-area independent lab sequence on oscilloscopes, function generators, meters, and a Keithley source-measure unit, comparing results with LTspice and QUCS simulation.
- Measured five 100 Ω resistors within their ±5% tolerance, RC rise and fall times of 11.1 ms that scaled with capacitance, and an LC resonance peak near 1.6 kHz compared with simulation.
- Measured multimeter loading on a high-impedance divider, then measured the divider through a unity-gain op-amp buffer; built and measured RF tank circuits on a proto-board and designed LC low- and high-pass RF filters.
- Measured photodiode I-V curves in dark, ambient, and laser light, LED optical power, and a laser diode power-current curve showing threshold.

PROJECTS

Used-Truck Deal Finder - Daily Listing Pipeline and Price Model

Oct 2026 - Present

Independent project for an Erie-area dealership

- Built a daily pipeline for an Erie-area dealership's used-truck buyer that prices private-party listings against area dealer retail and emails a reviewed PDF packet of underpriced trucks from a Linux server (trial running).
- Pulled about 1,300 used vehicles a day from 11 dealer websites' inventory endpoints and scraped about 450 by-owner listings from Craigslist's search feed and logged-out Facebook Marketplace with Playwright.
- Wrote negation-aware text rules for dealer, curbstoner, branded-title, and mechanical red flags, each fix backed by a regression test built from the real ad; the first full run removed 199 of 458 listings as dealer posts.
- Trained a five-seed gradient-boosted price model with a target-encoded make and model and a trim tier: median error about 9% in 5-fold cross-validation, down from 13% with plain categorical encoding.
- Measured error by price band (about 7% at $20,000 to $35,000, about 30% under $10,000) and set the pick rules to match: estimate $8,000 to $45,000, two or more close dealer comps, and a spread of at least $2,500 and 15%.

Superconductor Critical Temperature Modeling

Mar 2026

- Trained a Random Forest regressor predicting critical temperature for 21,263 superconducting compounds: test R^2 0.930 (RMSE 8.97 K), 5-fold cross-validated R^2 0.926.
- Showed with feature importance that thermal-conductivity descriptors drove the prediction, and applied PCA and k-means clustering to isolate regions containing high-temperature superconductors.

Image Classification with Convolutional Neural Networks

Jan 2026 - Mar 2026

DSCI 372M Machine Learning for Data Science, University of Oregon

- Trained a PyTorch convolutional neural network on 10-class FashionMNIST to 93.9% test accuracy, adding data augmentation after diagnosing memorization (coursework).
- Caught a starter-code bug that built the validation set from augmented training images, and implemented 2D convolution and max pooling from scratch in NumPy.

Calibrated Rare-Event Classifier (Expected Goals)

Jan 2026 - Mar 2026

DSCI 372M Machine Learning for Data Science, University of Oregon

- Trained a random forest classifier on soccer shot data and checked probability calibration (test expected calibration error 0.041, accuracy 69%).
- Found that training on class-balanced data inflated predicted goals almost fourfold (0.448 vs 0.122 per shot), corrected it by reweighting to the true 11.3% base rate, and confirmed the fix against a closed-form prior-shift correction and a Poisson-binomial test of season totals.

Machine Learning Algorithms from Scratch Jan 2026 - Mar 2026

DSCI 372M Machine Learning for Data Science, University of Oregon

- Implemented gradient descent and closed-form regression, a perceptron, a manual backpropagation pass, an entropy-based decision tree with cost-complexity pruning, k-means, hierarchical and spectral clustering, and PCA by eigendecomposition in NumPy.

Logistic Regression from Scratch (Income Classification) Jan 2026 - Mar 2026

DSCI 372M Machine Learning for Data Science, University of Oregon

- Implemented logistic regression with batch gradient ascent in NumPy on about 32,500 census records after mode imputation, z-score outlier handling, and one-hot encoding; test ROC AUC about 0.89.

Lake Erie Weather-Buoy Safety Model Nov 2024

- Merged 61,448 Lake Erie buoy observations (20-minute cadence, 2014 to 2021) with Erie airport weather and predicted wave height with ridge regression and PCA, using cross-validated feature selection and interaction terms; holdout RMSE about 0.29 ft.
- Tested wave conditions by wind direction with ANOVA (F = 362).

Resident-Assistant Shift Scheduler Oct 2024

- Built a constraint-optimization model in Python with OR-Tools that schedules on-call shifts across 12 staff teams under 8+ fairness and coverage constraints.
- Cut schedule building to seconds, an estimated 750+ administrative hours saved per year over manual scheduling.

Google Merchandise Store Analysis Mar 2025

- Analyzed 900,000+ e-commerce sessions in SQL on BigQuery and modeled purchase propensity with logistic regression.

FishTracker App Sep 2025

- Built an offline-first mobile app in Python and Kivy with a SQLite data model linking catches to GPS location and NOAA conditions.

Compound Defect-Count Model (Probability and Simulation) Apr 2025 - Jun 2025

DSCI 345M Probability and Statistics for Data Science, University of Oregon

- Modeled daily critical defects as a binomial count over a Poisson number of inspections, derived the expected count of 2 by iterated expectation, and confirmed it by Monte Carlo simulation (2.058 over 1,000 draws).
- Evaluated a gamma-Poisson marginal probability by numerical integration and a 1,000,000-draw Monte Carlo check (0.0529 vs 0.0525).

CAHOOTS Dispatch Share Analysis Apr 2024 - Jun 2024

DSCI 410L Data Science for Social Justice, University of Oregon

- Built a reproducible pandas pipeline over about 1.27 million Eugene CAD dispatch rows (2016 to 2023) and bootstrapped the share handled by CAHOOTS by staffing level: about 3.6% with one shift, 5.0% with two, and 8.7% with three.

TEACHING AND WORK EXPERIENCE

Learning Assistant - Applied Data Science for Social Justice Apr 2025 - Jun 2026

University of Oregon, Eugene, OR

- Led lab sessions and office hours across two academic years, guiding students through Python and pandas data cleaning, visualization, and analysis of CAHOOTS dispatch logs.
- Coached partner-facing presentations for a community stakeholder (White Bird Clinic).

Mathematics Paper Marker Mar 2025 - Jun 2026

University of Oregon, Eugene, OR

- Graded and gave written feedback across Linear Algebra, Mathematical Cryptography, and statistical modeling coursework; worked with instructors on rubric adherence and grading consistency.

Resident Assistant Sep 2023 - Jun 2026

University of Oregon Housing, Eugene, OR

- Conducted routine safety inspections, policy enforcement, and incident documentation for a 180+ resident community over a three-year appointment.
- Served as on-call first response for medical, safety, and facilities incidents, coordinating with professional staff and emergency services.
- Led community programming for residents.

Social Media Analytics and Marketing Intern Jun 2023 - Aug 2023

Humes, Waterford, PA

- Increased organic reach 117% in three weeks by analyzing engagement metrics and optimizing posting cadence in Meta Business Suite.

SKILLS

Cleanroom Processing: ISO 6 and ISO 4 cleanroom gowning and operations, photolithography (spin coat, bakes, mask alignment, direct write, develop, dose optimization), thermal oxidation, thermal evaporation and lift-off, wet etch, UV-ozone and O2 plasma cleaning, multilayer alignment, photomask layout (CleWin, KLayout), SU-8 molds and PDMS channels

Metrology and Characterization: stylus profilometry (Bruker Dektak), ellipsometry (Woollam), reflectometry (Filmetrics F20, F40-UV), optical microscopy, Keithley 2450 source-measure unit, semiconductor probe station, oscilloscopes, function generators, and photodetectors, free-space optical alignment and interferometry

Statistics and DOE: full factorial and fractional factorial designs, ANOVA and reduced models, residual diagnostics and Box-Cox transforms, Shapiro-Wilk, Levene, and Brown-Forsythe tests, Tukey HSD, Fisher LSD, and matched pairs, desirability optimization, regression (OLS, logistic, Poisson, ridge, lasso), cross-validation, bootstrap, censored maximum likelihood

Modeling and Machine Learning: discrete-event simulation, Markov chains and Poisson processes, phase-type service times, random forests and gradient-boosted trees, neural networks and CNNs (PyTorch), PCA and k-means clustering, constraint optimization (OR-Tools)

Programming and Tools: Python (pandas, NumPy, scikit-learn, PyTorch), R (tidyverse), SQL (incl. BigQuery), JMP, web scraping (Playwright), Excel, Tableau, Git and GitHub, Bash, LaTeX, QUCS and LTspice